

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.Tech AI/DS	Date:	

Code of Conduct

I Nirmal Kasi 192424248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nirmal

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility
- Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Q1. Reference Resolution and Discourse Coherence

Given paragraph:

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

a) Referring Expressions and Possible Antecedents

The referring expressions are **He, him, it, they,** and **the book**.

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	The book
they	Ravi and Arun
the book	The book mentioned earlier

b) Applying the Given Constraints

The system uses **gender and number agreement, recency, grammatical role, semantic compatibility, and discourse coherence** to resolve references.

1. “He”

Possible antecedents:

- Ravi
- Arun

Both Ravi and Arun are singular male entities, so **gender and number agreement alone cannot completely resolve the ambiguity**.

However, Ravi is the subject of the preceding sentence and is the person who is likely to be looking for a book.

Therefore:

He → Ravi

2. “him”

Possible antecedents:

- Ravi
- Arun

The sentence is:

“Arun helped him find it.”

Arun is already identified as the person performing the helping action. Therefore, **him** naturally refers to Ravi, the person receiving the help.

him → Ravi

3. “it”

Possible antecedent:

- Book

The sentence says:

“Arun helped him find it.”

A book is something that can be found, so semantic compatibility strongly supports:

it → the book

4. “they”

Possible antecedents:

- Ravi
- Arun

Because **they** is plural, it refers to both people:

they → Ravi + Arun

5. “the book”

This is a definite noun phrase referring back to the previously introduced book.

the book → previously mentioned book

c) Reference Resolution Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi, Arun	Ravi = Valid	Subject continuity and discourse coherence
He	Ravi, Arun	Arun = Possible	Gender and number agree, but weaker contextual interpretation
him	Ravi, Arun	Ravi = Valid	Arun is the helper, Ravi receives the help
him	Ravi, Arun	Arun = Invalid	Arun cannot naturally help himself in this context
it	Book	Valid	A book can be found
they	Ravi + Arun	Valid	Plural agreement
the book	Book	Valid	Refers to previously introduced entity

d) Final Coreference Chains

Person Chain 1:

Ravi → He → him

Person Chain 2:

Arun → Arun

Object Chain:

Book → it → the book

Plural Entity Chain:

Ravi + Arun → they

e) Rewritten Paragraph

“Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.”

This version removes the ambiguity created by the pronouns while preserving the original meaning.

f) Ambiguity if “He” Appeared Earlier

If the sentence **“He was looking for a book”** appeared before Arun was introduced, then the reference would be less ambiguous because only Ravi would be available as an antecedent.

After both Ravi and Arun are introduced, however, the pronoun **“He”** can potentially refer to either person because both satisfy the singular and male agreement constraints.

Therefore, additional information such as **recency, grammatical role, semantic compatibility, and discourse context** is needed to select the correct antecedent.

Q2. Dialog Act Recognition and Response Generation

Conversation:

Student: “I submitted my assignment, but I am worried that I may have made many mistakes.”

Required dialog acts: **Reassure + Advise.**

a) Required Dialog Acts

The required dialog acts are:

1. **Reassure**
The agent should reduce the student's worry and provide emotional reassurance.
2. **Advise**
The agent should provide useful guidance about reviewing mistakes and improving future work.

Therefore:

Student Concern → Reassurance + Advice

b) Important Entities

The important entities that must remain consistent throughout the response are:

- **Assignment**
- **Mistakes**
- **Student / You**

- **Feedback**
- **Improvement**

Maintaining these entities ensures that the generated response remains related to the student's original concern.

c) Three Possible Responses

Response 1:

“Don’t worry too much about the assignment; mistakes are a normal part of learning. You can review your work and use feedback to improve, which will help you become more confident.”

Response 2:

“It is normal to worry after submitting an assignment, but you can use any mistakes as opportunities to improve. Review the assignment and consider the feedback so you can become more confident in your future work.”

Response 3:

“Try not to worry too much about the mistakes because submitting the assignment is already an important step. Review your work later, use the feedback to improve, and focus on becoming more confident.”

All three responses contain reassurance and advice while maintaining the required entities and a positive tone.

d) Comparison of Responses

Criterion	Response 1	Response 2	Response 3
Coherence	High	High	High
Constraint Satisfaction	High	High	High
Politeness	High	High	High
Relevance	High	High	High
Reassurance	Strong	Strong	Strong
Advice	Strong	Strong	Strong

e) Best Response

Response 2 is the best response.

Reasons:

- It directly addresses the student's worry.
- It provides reassurance without dismissing the student's concern.
- It advises the student to use mistakes as learning opportunities.
- It maintains the entities **assignment**, **mistakes**, **feedback**, and **improve**.
- It maintains a positive and polite tone.
- It does not provide contradictory advice.

Therefore, Response 2 provides the best combination of **coherence**, **constraint satisfaction**, **politeness**, and **relevance**.

f) Effect of Losing Entity Coherence

If entity coherence is not maintained, the response may stop referring correctly to the student's assignment or mistakes. The agent could produce an unrelated response or confuse the entities involved.

For example, if the system suddenly discusses a different task instead of the submitted assignment, the conversation becomes inconsistent and the response becomes less useful.

Therefore, maintaining entity coherence is essential for generating relevant and logically connected conversational responses.

Q3. Word Sense Disambiguation and Meaning Representation

Source sentence:

“Priya sat on the bank and watched the boats moving across the water.”

a) Possible Meanings of “Bank”

The word **bank** has two possible meanings in the given question:

1. **Financial institution**
2. **Side of a river**

b) Correct Sense

The correct sense is:

Bank = Side of a river

The contextual clues clearly indicate the river-related meaning.

Evidence:

- Priya **sat on the bank**.
- She watched **boats**.
- The boats were moving across the **water**.

Boats and water are strongly associated with a river environment. A financial institution does not fit this context.

Therefore:

bank → river bank

This is an example of **Word Sense Disambiguation**, where the correct meaning of an ambiguous word is selected using its surrounding context.

c) Predicate Logic Representation

Let:

- p = Priya
- b = river bank

- $z = \text{boat}$
- $w = \text{water}$

Suitable predicates are:

- $\text{Sit}(x) = x \text{ sits}$
- $\text{Bank}(y) = y \text{ is a river bank}$
- $\text{Beside}(x, y) = x \text{ is beside } y$
- $\text{Watch}(x, z) = x \text{ watches } z$
- $\text{Boat}(z) = z \text{ is a boat}$
- $\text{Move}(z) = z \text{ moves}$
- $\text{Water}(w) = w \text{ is water}$

Predicate representation:

$\text{Sit}(\text{Priya}) \wedge \text{Bank}(b) \wedge \text{Beside}(\text{Priya}, b) \wedge \text{Watch}(\text{Priya}, z) \wedge \text{Boat}(z) \wedge \text{Move}(z) \wedge \text{Water}(w)$

A more relationship-focused representation is:

$\text{Sit}(\text{Priya})$

$\text{Bank}(b)$

$\text{Beside}(\text{Priya}, b)$

$\text{Boat}(z)$

$\text{Water}(w)$

$\text{Move}(z, w)$

$\text{Watch}(\text{Priya}, z)$

This represents the important entities and relationships in the sentence.

d) Unambiguous English Paraphrase

“Priya sat beside the river and watched the boats moving across the water.”

This paraphrase removes the ambiguity of the word **bank** by explicitly saying **beside the river**.

e) WSD in Machine Translation

Word Sense Disambiguation improves Machine Translation by identifying the correct meaning of an ambiguous word before translation.

For example:

Original:

Priya sat on the bank.

If **bank** is incorrectly interpreted as a financial institution, the translated sentence will have the wrong meaning.

If WSD identifies:

bank → side of a river

the translation preserves the correct semantic meaning.

Therefore, WSD helps Machine Translation by:

1. Identifying ambiguous words.
2. Examining their context.
3. Selecting the correct sense.
4. Preserving semantic relationships.
5. Producing a more accurate and coherent translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	3	4	4	4	19	
2	4	4	4	3	4	19	
3	4	4	4	4	4	20	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____